the truly best strategy, in a more general and less arbitrary sense? Readers of earlier chapters will already be prepared to find the answer in the theory of evolutionarily stable strategies.

I was one of those to whom Axelrod circulated his early results, with an invitation to submit a strategy for the second tournament. I didn't do so, but I did make another suggestion. Axelrod had already begun to think in ESS terms, but I felt that this tendency was so important that I wrote to him suggesting that he should get in touch with W. D. Hamilton, who was then, though Axelrod didn't know it, in a different department of the same university, the University of Michigan. He did indeed immediately contact Hamilton, and the result of their subsequent collaboration was a brilliant joint paper published in the journal *Science* in 1981, a paper that won the Newcomb Cleveland Prize of the American Association for the Advancement of Science. In addition to discussing some delightfully way-out biological examples of Iterated Prisoner's Dilemmas, Axelrod and Hamilton gave what I regard as due recognition to the ESS approach.

Contrast the ESS approach with the 'round-robin' system that Axelrod's two tournaments followed. A round-robin is like a football league. Each strategy was matched against each other strategy an equal number of times. The final score of a strategy was the sum of the points it gained against all the other strategies. To be successful in a round-robin tournament, therefore, a strategy has to be a good competitor against all the other strategies that people *happen* to have submitted. Axelrod's name for a strategy that is good against a wide variety of other strategies is 'robust'. Tit for Tat turned out to be a robust strategy. But the set of strategies that people happen to have submitted is an arbitrary set. This was the point that worried us above. It just so happened that in Axelrod's original tournament about half the entries were nice. Tit for Tat won in this climate, and Tit for Two Tats would have won in this climate if it had been submitted. But suppose that nearly all the entries had just happened to be nasty. This could very easily have occurred. After all, 6 out of the 14 strategies submitted were nasty. If 13 of them had been nasty, Tit for Tat wouldn't have won. The 'climate' would have been wrong for it. Not only the money won, but the rank order of success among strategies depends upon which strategies happen to have been submitted; depends, in other words, upon something as arbitrary as human whim. How can we reduce this arbitrariness? By 'thinking ESS'.